

Maurice D. Hanisch

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Education

- California Institute of Technology, Ph.D. Math & CS** 09/2025 – Present
Pasadena, USA
- Pursuing research in ML for quantum chemistry and materials science, advised by Prof. Anima Anandkumar.
 - Fully funded through the Kortschak Fellowship, awarded based on academic excellence and research potential.
- ETH Zurich, M.Sc. Physics** 5.77/6.00 (Top 12%) 09/2022 – 05/2024
Zurich, Switzerland
- Expanded analytical skills with a focus on quantum theory and information technologies, with 30% excess credits in mathematics, ML and probabilistic AI involving 8 team projects, all graded 6.00/6.00 (🟢 projects).
- LMU Munich, B.Sc. Physics** 1.28/1.00 (Top 10%) 10/2018 – 09/2022
Munich, Germany
- Acquired extensive grounding in key physics fields, providing a strong foundation in the principles driving modern technologies, and fostering advanced analytical and problem-solving skills.

Work Experience

- Bloomberg CTO Office, Security Research Intern, Julien Vanegue's Team** 06/2025 – 09/2025
New York, USA
- Stress-tested the CHERI CPU architecture, a \$300 million industry-academic security effort, by attempting to break its memory-safety guarantees to determine its suitability for Bloomberg's infrastructure.
 - Discovered multiple attack vectors that compromise both spatial and temporal memory safety guarantees.
 - Contributed mitigations to the open-source CHERI software via multiple PRs (🟢 jemalloc, 🟢 dmalloc, 🟢 c-guide).
 - Co-authored a paper accepted at ACM CCS 2026.
- California Institute of Technology, ML Research Intern, Anima Anandkumar's Group** 09/2024 – 05/2025
Pasadena, USA
- Led a large-scale benchmark, retraining seven SOTA molecular models for open-shell and charged systems, validating our proposed method, *OrbitAll*, in our published work.
 - Developed a differentiable quantum feature generation pipeline for GNN-based models, enabling end-to-end differentiability of physics-informed learning of quantum chemical properties.
 - Submitted PRs to chemistry ML tooling: PyTorch (🟢 optimized eigensolver selection) and tblite (🟢 lightweight SCF restarts).

Research Experience

- IBM, Master's Thesis, Stefan Woerner's Group** | 📄 Thesis (PDF), 🌐 Project repository 6.00/6.00 09/2023 – 05/2024
Zurich, Switzerland
- Conducted state-of-the-art research on error correction at a global leader in the quantum computing industry.
 - Developed a Python/C++ decoding pipeline that scaled to billions of measurement records per run, enabling one of IBM's then-largest error-correction experiments.
 - Designed and deployed the full experimental pipeline on IBM Q hardware, showing that analog-information decoding can improve logical error rates by up to 30x in large-scale repetition-code experiments.
- ETH Zurich, Summer Project, Jonathan Home's Group** | 📄 Thesis (PDF) 06/2023 – 08/2023
Zurich, Switzerland
- Investigated the motional interaction of ions trapped in an electromagnetic field for use in quantum computing.
 - Designed and simulated trapping potentials using Python to enable a beamsplitter interaction.
 - Experimentally validated the designed potentials on GKP-encoded qubits within a trapped-ion setup.
- MPI for Quantum Optics, Bachelor's Thesis, Ignacio Cirac's Group** | 📄 Thesis (PDF) 1.30/1.00 04/2022 – 09/2022
Garching, Germany
- Investigated efficient algorithm design and resource optimization for Gaussian quantum circuits.
 - Applied complexity theory and mathematical techniques to prove that bosonic Gaussian circuits can be implemented on quantum computers with exponentially fewer qubits compared to a direct encoding.

Selected Publications

- Learning the Kohn-Sham map with neural operators for quasi-linear scaling density functional theory**
○ D. Khan*, M.D. Hanisch*, N. Argatoff, E. Xie, S. Sharma, A. Anandkumar - arXiv:2608.23895 (2026)
- Soft information decoding with superconducting qubits**
○ M.D. Hanisch, B. Hetényi, J.R. Wootton - APS Open Sci. 1, 000019 (2026)
- MGB: The Material Generation Benchmark**
○ L. Yan, B.S. Kang, M.D. Hanisch, J. Ma, A. Anandkumar - NeurIPS AI4Mat Workshop (2025)
- OrbitAll: A Unified Quantum Mechanical Representation Deep Learning Framework for All Molecular Systems**
○ B.S. Kang, V.C. Bhethanabotla, A. Tavakoli, M.D. Hanisch, ..., A. Anandkumar - arXiv:2507.03853 (2025)

Extracurricular Activities and Competitions

- Participated in 10+ technical competitions and workshops across cities like Paris, Munich, Zurich and New York, sponsored by companies ranging from consulting firms (Bain, McKinsey) to tech giants (Bloomberg, Infineon, NVIDIA, EDF).
 - Frequently selected through competitive application processes, often with expenses (travel, accommodation, meals) covered.
 - Won over \$3500 in hackathon prizes for projects using recurrent NNs, time series, LLMs, CUDA-Q, SageMaker and Hugging Face.
- Hackathons:** 1st/30 at ETH QHack 2024: 🟢 NVIDIA task | 1st/12 at LOQC 2023: 🟢 EDF task | 2nd/6 at ETH QHack 2023: 🟢 IQM task

Technical Skills and Interests

Programming: Python, C, C++, Fortran, Matlab, SQL | PyTorch, TensorFlow, Scikit-learn | Git, Linux, Latex, Docker
Languages: German (native), French (native), English (C1: IELTS 8/9), Italian (A1)
Interests: Weightlifting, Volleyball, Calisthenics, Surfing, Chess